Estimating Power-Law Exponents from Remote Sensing Data

Overview

This vignette serves two primary purposes:

Empirical Validation: We compare the package's estimates of the power-law exponent (α) and total stem abundance (N_{tot}) against field-measured data from the National Ecological Observatory Network (NEON).

Performance Benchmarking: We track execution times for the key computational steps—breakpoint detection, data truncation, and Bayesian inference—to demonstrate the workflow's scalability across multiple sites.

Multi-Site Alpha Estimation

We first define the set of sites to process:

```
sitevector <- c("ABBY", "BART", "BLAN", "BONA", "CLBJ", "DELA", "GUAN",
                "HARV", "JERC", "LENO", "MLBS", "OSBS",
                "RMNP", "SCBI", "SERC", "SJER", "SOAP", "TALL", "TEAK",
                "WREF")

results_model <- list()
results_summary <- data.frame()
```

We iterate over each site, performing breakpoint detection, filtering, and alpha estimation.

```
# Initialize outputs
results_model <- list()
results_summary <- data.frame()
runtime_table <- data.frame(
  siteID = character(),
  potential_break_time = numeric(),
  truncate_filter_time = numeric(),
  fit_alpha_model_time = numeric()
)

for(site in sitevector){

  site_data <- neon_data %>%
    filter(siteID == site & dbh <= 50) %>%
    data.frame()

  # Skip sites with insufficient data
```

```r
if(nrow(site_data %>% filter(dbh > 10)) < 25) next

# Step 1: Identify potential breakpoint
t1 <- system.time({
  breakpoint_kde <- potential_break(
    data = site_data,
    n_bootstrap = 1000,
    trim_max = 50
  )
})

# Step 2: Filter and truncate data based on breakpoint
t2 <- system.time({
  filtered_data <- truncate_filter(
    breakpoint_kde_results = breakpoint_kde,
    min_size = 10
  )
})

# Step 3: Fit Bayesian alpha model
site_priors <- alpha_priors %>% filter(siteID == site)

LAI_value <- Leaf_area_index %>%
  rename(siteID = site) %>%
  filter(siteID == site) %>%
  pull(Leaf_area_index)

t3 <- system.time({
  fit <- fit_alpha_model(
    filtered_data= filtered_data,
    LAI = LAI_value,
    prior_mean = site_priors$prior_mean,
    prior_sd = site_priors$prior_sd,
    iter = 9000,
    warmup = 6000,
    chains = 4,
    cores = 4
  )
})

# Store results
results_model[[site]] <- fit
results_summary <- rbind(
results_summary,
fit$posterior_summary %>%
mutate(siteID = site)
)
# Record runtimes
runtime_table <- rbind(
  runtime_table,
  data.frame(
```

```
      siteID = site,
      potential_break_time = t1["elapsed"],
      truncate_filter_time = t2["elapsed"],
      fit_alpha_model_time = t3["elapsed"]
    )
  )
}
```

```
## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1
```

```
## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
```

```
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
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## compilation terminated.
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library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
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library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
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library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
```

```
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library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
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library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
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library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
```

```
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
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library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
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## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
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library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
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##   679 | #include <cmath>
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```
##       |          ^~~~~~~
## compilation terminated.
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## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
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library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
```

```
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
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library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~
```

```
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
```

```
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~
```

```
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
```

```
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~
```

```
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
```

```
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~
```

```
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
```

```
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1
```

```
# Optionally, calculate summary statistics
runtime_summary <- runtime_table %>%
  summarise(
    mean_break = mean(potential_break_time),
    mean_truncate = mean(truncate_filter_time),
    mean_fit = mean(fit_alpha_model_time),
    median_break = median(potential_break_time),
    median_truncate = median(truncate_filter_time),
    median_fit = median(fit_alpha_model_time),
    max_fit = max(fit_alpha_model_time),
    min_fit =min(fit_alpha_model_time)
  )

runtime_summary
```

```
##   mean_break mean_truncate mean_fit median_break median_truncate median_fit
## 1     1.3955         0.181  92.3465         1.34           0.135     73.945
##   max_fit min_fit
## 1  214.17      60
```

Visualizing Alpha Estimates

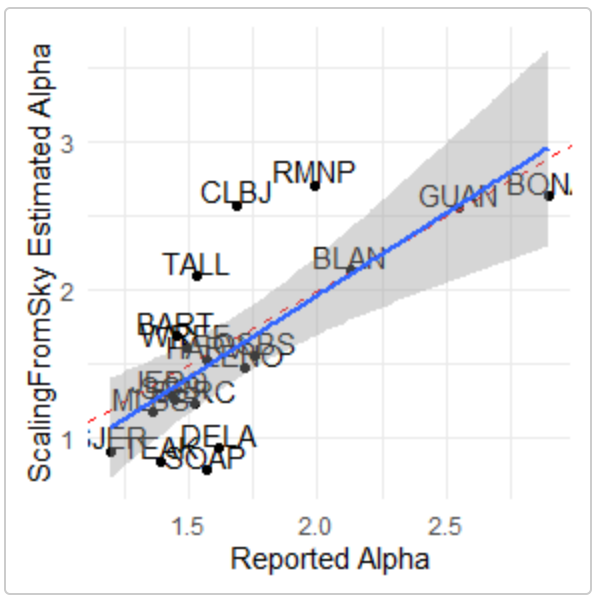
We can quickly compare estimated alpha values to previously reported values for validation. The 1:1 line in red would represent a “perfect” replication of measured data by our interpolation. Notice that we add a filter to remove sites with low Bayesian R2 values, as such values would suggest that the fit of the Bayesian distribution to find (α) was poor for that site. In this case, the site “ABBY” was removed from the analysis:

```r
data("neon_truemeasuredalpha")

ggplot(results_summary%>%
         inner_join(neon_truemeasuredalpha)%>%
         filter(R2_kernel > 0.6), aes(x = alpha, y = mean, label = siteID)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
  geom_text(nudge_y = 0.1) +
  labs(x = "Reported Alpha", y = "ScalingFromSky Estimated Alpha") +
  theme_minimal() +
  geom_smooth(method = "lm")
```

```
## Joining with `by = join_by(siteID)`
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label.
## ℹ This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## ℹ Did you forget to specify a `group` aesthetic or to convert a numerical
##   variable into a factor?
```



Notes

- The workflow uses **bootstrap KDEs** to identify size distribution peaks and potential breakpoints.
- Filtering ensures the model fits to the **upper tail**, where Pareto behavior is expected.
- LAI and breakpoint corrections account for **reduced detectability of small trees** in remote sensing data.
- Sites with fewer than 25 trees after filtering are skipped to ensure robust estimation.

Estimation of Total Stem Abundance (N_tot)

Following estimation of the size–abundance scaling parameter (alpha) for each site, the next stage in the ScalingFromSky workflow is estimation of **total stem abundance (N_tot)**. This step scales the relative size–abundance information inferred from remote sensing to an absolute estimate of tree density at each site.

This is implemented using the `estimate_total_trees()` function, which propagates uncertainty from the previously fitted alpha model into posterior estimates of total abundance. The model uses weakly informative priors on total abundance and leverages the fitted size–abundance structure to infer the number of trees consistent with the observed size distribution.

Estimating total abundance across sites

```
# Storage objects
# Initialize storage
results_abundance_model <- list()
results_abundance_summary <- data.frame()


abundance_runtime <- data.frame(
siteID = character(),
elapsed_sec = numeric(),
stringsAsFactors = FALSE
)

for (site in names(results_model[1])) {


message(paste("Estimating total abundance for site:", site))


# Extract fitted alpha model for this site
alpha_fit <- results_model[[site]]


# Time the abundance estimation
t_start <- Sys.time()


trees <- estimate_total_trees(
alpha_model_output = alpha_fit,
N_tot_prior_mean = 1250,
N_tot_prior_sd = 625,
warmup = 6000,
iter = 9000,
siteID = site
)


t_end <- Sys.time()
```

```
# Store full model output
results_abundance_model[[site]] <- trees


# Store posterior summary for N_tot
if (!is.null(trees$posterior_summary)) {
results_abundance_summary <- rbind(
results_abundance_summary,
trees$posterior_summary
)
}


# Store runtime
abundance_runtime <- rbind(
abundance_runtime,
data.frame(
siteID = site,
elapsed_sec = as.numeric(difftime(t_end, t_start, units = "secs"))
)
)
}


## Estimating total abundance for site: ABBY


## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.5.0 from
## https://cran.r-project.org/bin/windows/Rtools/.


## Trying to compile a simple C file


## Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 14.2.0'
## gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/unsupported"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/BH/include" -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/src/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -
I"C:/Users/adam.local/AppData/Local/R/win-library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -
DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
"C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -I"C:/rtools45/x86_64-
```

```
w64-mingw32.static.posix/include"      -O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c
foo.c -o foo.o
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from C:/Users/adam.local/AppData/Local/R/win-
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22,
##                  from <command-line>:
## C:/Users/adam.local/AppData/Local/R/win-
library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal error: cmath: No such file
or directory
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1
```

```
## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.5.0 from
## https://cran.r-project.org/bin/windows/Rtools/.
```

```
# Per-site runtimes
abundance_runtime
```

```
##   siteID elapsed_sec
## 1   ABBY     161.028
```

```
# Summary statistics
abundance_runtime_summary <- abundance_runtime %>%
summarise(
n_sites = n(),
mean_sec = mean(elapsed_sec),
median_sec = median(elapsed_sec),
sd_sec = sd(elapsed_sec),
min_sec = min(elapsed_sec),
max_sec = max(elapsed_sec)
)
```

```
abundance_runtime_summary
```

```
##   n_sites mean_sec median_sec sd_sec min_sec max_sec
## 1       1  161.028    161.028     NA 161.028 161.028
```

Interpretation

The resulting posterior distributions of N_tot represent site-level estimates of total stem abundance, conditional on the inferred size–abundance scaling and prior information on plausible stand densities. These estimates can be used directly for site-to-site comparisons, downstream modeling, or integration with forest demographic and carbon accounting workflows.

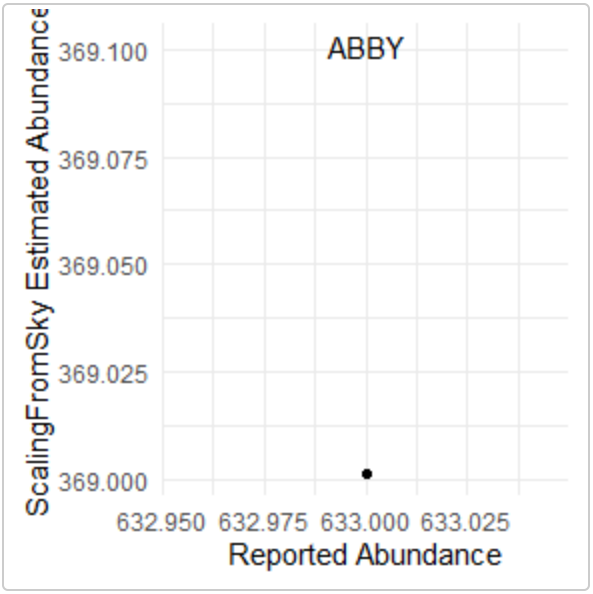
Visualizing Abundance Estimates

We can quickly compare estimated abundance values to previously reported values for validation. The 1:1 line in red would represent a “perfect” replication of measured data by our interpolation:

```
#Real world abundance data
abundancedata <- data.frame(
  site = c(
    "ABBY","BART","BLAN","BONA","CLBJ","DEJU","DELA","DSNY","GUAN",
    "HARV","HEAL","JERC","KONZ","LAJA","LENO","MLBS","MOAB","NIWO",
    "OSBS","RMNP","SCBI","SERC","SJER","SOAP","SRER","TALL","TEAK",
    "WREF","YELL"
  ),
  abundance = c(
    633,1965,251,1576,629,984,1269,48,586,
    2111,47,294,40,8,740,1176,42,1042,
    869,1307,741,1055,110,1044,8,1084,1033,
    1225,452
  )
)

ggplot(results_abundance_summary%>%
         inner_join(abundancedata), aes(x = abundance, y = mean, label = site)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
  geom_text(nudge_y = 0.1) +
  labs(x = "Reported Abundance", y = "ScalingFromSky Estimated Abundance") +
  theme_minimal() +
  geom_smooth(method = "lm")
```

```
## Joining with `by = join_by(site)`
## `geom_smooth()` using formula = 'y ~ x'
```


ABBY
ScalingFromSky Estimated Abundance
369.100
369.075
369.050
369.025
369.000
632.950
632.975
633.000
633.025
Reported Abundance